



CropScan AI Sentinel

CAB V4.2

Oak Ridge Commercial Farm • 1,850 Total Acres

Active Sector: 4B (140 Acres Russet Potatoes)

19.4°C | 88% Humidity | Wind: 6.2 mph NW

[SEVERE SPORE VELOCITY]

[AGENT ACTIVE]



1. Drone Ingestion

✓
Tile #4410 •
Orthomosaic Sync



2. CV Pathology ✓

Late Blight
Confirmed (96.4%)



3. Weather Risk ✓

72h Spore
Velocity High



4. EPA Formulation

✓
Chlorothalonil
720 SC @ 1.5
pt/ac



5. Tractor Dispatch

✓
Work-Order #W0-
0941 Compiled

[AUTONOMOUS CYCLE COMPLETED (2.4S)]



Computer Vision Leaf Inspector

[Curated Samples]

[Upload Field Image]

[Live Drone Stream]



• Phytophthora infestans (Late Blight) • 96.4%

Secondary Necrosis • 18.5% Area

SENSOR: MULTISPECTRAL-CAM #04

RESOL: 4096x2160 • NDVI: 0.28 (STRESSED)

✓ Potato Late Blight (Selected)

Tomato Early Blight

Corn Common Rust

Healthy Control

PATHOGEN CLASSIFICATION

Potato Late Blight

Phytophthora infestans

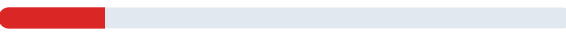
BIOSAFETY THREAT

[CRITICAL TIER 3]

Spread Index: 8.8/10

CANOPY DEGRADATION

18.5% Affected Surface Area



Deterministic EPA Tank-Mix

AUTO-CALC

Chlorothalonil 720 SC

EPA Reg. #50534-188

Broad-spectrum contact fungicide, Frac Group M05.

Application Rate: 1.5 pt / acre in 20.0 gal water
Target Carrier Volume: 2,800 Gallons Total Solution
Concentrate Needed: 26.2 Gallons (\$2,140 input cost)

✓ Wind: 6.2 mph (<10 mph threshold). Buffer: 100ft verified.



Tractor Work-Order Dispatch

STATION DISPATCHED

Ticket: #W0-2026-0941

Target: Sector 4B (140 Acres)

ASSIGNED UNIT

John Deere R4045 #02

BOOM WIDTH

120 ft (Pulse-Width Mod)

OPERATOR

Telemetry Autonav #12

NOZZLE PSI

42 PSI • Coarse Droplets

Export Work-Order PDF

Dispatch to Tractor



Financial & Environmental ROI

REAL-TIME VALUATION

PROTECTED YIELD

\$32,400

Basis: \$14.50/cwt Russet

MICRO-TARGETING

34% Saved

vs. Blanket Broadcast Spray

CARBON OFFSET: 142 kg CO2e

GROUNDWATER BUFFER: COMPLIANT